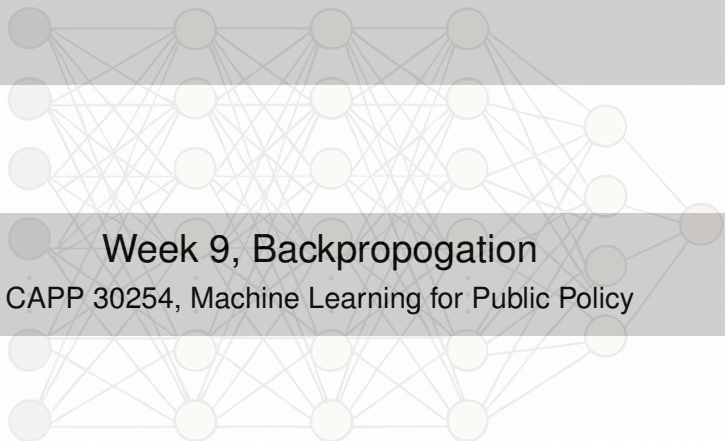




Week 9, Backpropagation

CAPP 30254, Machine Learning for Public Policy

CAPP 30254
MACHINE LEARNING
FOR PUBLIC POLICY

Neural networks

Reminder...

Neural networks are inspired by the brain's biological process. The brain is made of neurons and neurons receive signals, process them, and pass them on to another neuron.

Neural networks power many modern AI systems. They can learn complex, nonlinear patterns in ways that modern linear models can't achieve.

for CAPS students

In particular,[✓] neural networks are commonly used to find patterns, make predictions, and make decisions in public policy.

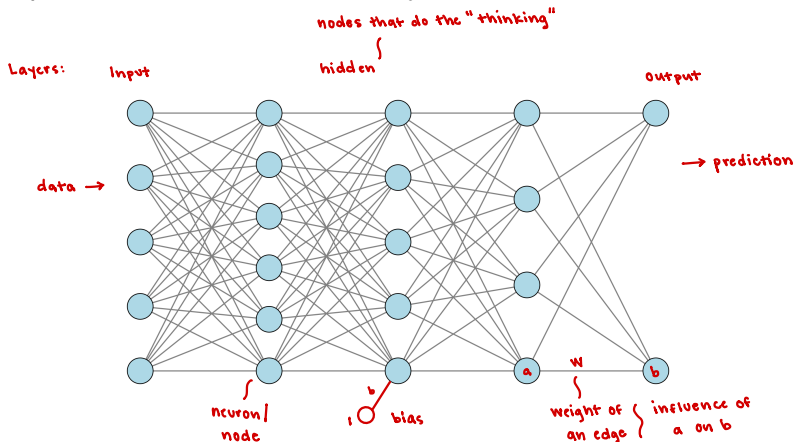
- Detect illegal deforestation or urban heat islands from satellite images
- Determine which neighborhoods are at risk of gentrification

Anatomy of a neural network

σ : activation function

$\mathcal{L}$: loss function

Consider the following neural network with four input nodes, an output node, and three hidden layers.



Forward and backward



learns w 's

A model learns its weights by passing information forward and backward through layers of connected nodes, minimizing a loss function just like other models we've seen.

minimize l

Passing data forward through the network is called a *forward pass* or *forward propagation*.

After the forward pass, the error (loss) is calculated and is propagated backward through the network. Stochastic gradient descent is used to find the weights that minimize the loss function. Moving backwards through the network is called *backpropagation*. Each backward pass changes the edge weights a tiny bit. After enough back-and-forth, the model will eventually make good predictions.

Common pitfalls



When gradients are backpropogated through the network layers, they might become very small or very large.

When gradients become very small, called *vanishing gradients*, layers close to the input might learn very slowly (or not at all).

On the other hand, when gradients become very big (*exploding gradients*) weights become too big and training diverges.

What other pitfalls might you encounter when training a neural network?

Fix:

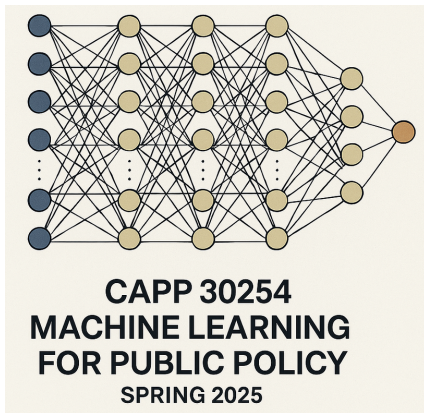
- normalise input layer
- better weight initialization
- gradient clipping

$$\nabla = \nabla / \|\nabla\|$$

Vanishing: use ReLU/leaky ReLU

Exploding: smaller learning rate

$$\eta = \eta / 10$$



1

¹Generated by GPT-4, DALL-E 3 after the prompt: “Could you make a complex neural network with ‘CAPP 30254 Machine Learning for Public Policy, Spring 2025’”